

Solution Requirements

Date	15 March 2026 26 June 2026
Team ID	XXXXXXX EduGenie - Google Gemini Powered Learning Assistant
Project Name	XXXXXXX EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>[Describe how users will</i> Single user profile initialized locally via settings; no <i>securely log in or verify their identity]</i>	user auth required for demo. Low
2	Authorization levels	All users have equal access to ask, explain, quiz, s <i>[Define the user roles and what access/permissions each role has]</i>	ummarize, and recommend modules. Low
3	External interfaces	Connects to Google Gemini API via HTTPS client; <i>[Detail any third-party APIs, external databases, or</i>	falls back to local HuggingFace T5 High

		<i>services the system must connect to]</i>	
4	Transactions processing	<i>[Explain how data</i> Queries and AI responses are processed as async <i>exchanges, payments, or core system transactions will be handled]</i>	requests and logged to JSON files. High
5	Reporting	<i>[Describe the types of</i> Interactive browser dashboards displaying query hi roadmaps. roadmaps. <i>reports, dashboards, or data exports the system must generate]</i>	story, quiz scores, and learning
6	Business rules, etc.	Gemini responses must match expected JSON sch <i>[List specific logic, parsing fails. constraints, or workflows unique to the business process]</i>	emas; fallback triggers automatically if High
7	Compliance to laws or regulations	No personal data stored; logs are stored locally in d <i>[Specify any legal, industry, privacy. or data privacy standards the system must adhere to (e.g., GDPR, HIPAA)]</i>	ata/*.json files, complying with basic Low
8	<i>Other</i>	Light/dark theme toggle state persists in localStora <i>[Add any additional functional requirements here]</i>	ge across browser sessions.

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	FastAPI routing response overhead < 10 ms; Gemini <i>[Describe response times, API latency depends on network. API latency depends on network. processing speed, etc.]</i>	(e.g., Page loads in < 2 seconds) Page load time < 2s
2	Scalability	<i>[Describe how the system</i> FastAPI <i>handles high concurrent request volume using async event loop. async event loop. handles growth in users or data]</i>	(e.g., Supports up to Up to 100 concurrent requests 10,000 concurrent users)
3	Security & Data Privacy	<i>[Detail data protection, API keys loaded securely via environment variables; input fields sanitized. input fields sanitized. encryption, and vulnerability management]</i>	(e.g., End-to-end Zero hardcoded secrets AES-256 encryption)
4	Reliability & Availability	<i>[Specify uptime requirements</i> Dynamic fallback to local offline T5 model / mock mode if Gemini API key is missing. <i>and fault tolerance]</i>	100% request resilience (e.g., 99.9% uptime per month)
5	Usability & Accessibility	Responsive, glassmorphic layout optimized for mobile and desktop screens. <i>[Describe user experience and standards and accessibility compliance]</i>	Intuitive single-click navigation. (e.g., Complies with WCAG 2.1 AA standards)
6	Other	<i>[Add any additional</i> Modular Python <i>codebase following clear clean architecture guidelines. architecture guidelines. non-functional requirements (e.g., Maintainability, Portability)]</i>	Code readability score > 90%